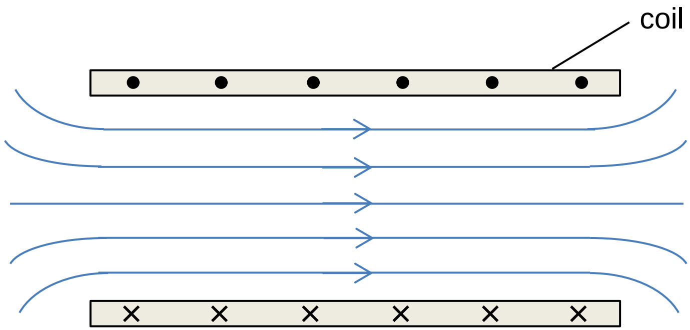
7	
(a)(i)	
	Correct direction of magnetic field with at least 5 lines
	Correct shape with equal distance between lines inside the coil Lines must be straight inside the coil Lines must curve outside the coil
(a)(ii)	To reduce resistance of the coil to allow for higher current to be used in coil OR With high current, resistance of the coil increases, cooling will prevent overheating.

(b)	Scanner generates a very strong magnetic field causes a strong magnetic force to attract metallic implants. This may harm patients with these implants by dislodging the metallic parts.
(c)(i)	The hydrogen nuclei has one unpaired proton.
	Hence its net spin is 1/2.
(c)(ii)	The carbon nuclei has 3 pairs of protons and 3 pairs of neutrons. Hence, it has no net spin.
	It is unable to absorb photons of a particular frequency.
(d)	Energy of photon $= hf$ $= h\nu$ $= h\gamma B$ $= (6.63\times10^{-34})(42.58\times10^{6})(1.5)$ $= 4.235\times10^{-26}$ J
	$= \dfrac{4.235\times10^{-26}}{1.60\times10^{-19}}$ eV
	$= 2.65\times10^{-7}$ eV (3 s.f.)
(e)(i)	Damage or destroy living cells OR Break bonds that hold water molecules together, forming toxic substances OR Creating free radicals
(e)(ii)	Energy of X-ray photon $= \dfrac{hc}{\lambda}$ $= \dfrac{(6.63\times10^{-34})(3.00\times10^{8})}{10^{-10}}$
	$= 1.99\times10^{-15}$ J (3 s.f.) Since energy of X-ray photon > 6.0×10^{-19} J, X-rays are ionizing.
(f)(i)	Phosphorus does not have an isotope.
(f)(ii)	The hydrogen ${}^{1}_{1}\text{H}$ nuclei has a higher natural abundance of 0.99985 and is the most common hydrogen isotope.
	It is also has the highest biological abundance of 0.63, which makes it the most common element in the human body.
(f)(iii)	Soft tissues are mostly made up of water and hydrogen nuclei are naturally found in water.
	When an external magnetic field is applied, the hydrogen nuclei can absorb photons of a particular frequency.
	As the nuclei return to their resting alignment, energy is emitted and converted to images of the soft tissues.